AI-Powered Multi-Brand E-Bike Recommendation and Smart Monitoring System

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Abstract — The rapid growth and diversity of the electric bicycle (e-bike) market makes model selection difficult for non-technical buyers, who must otherwise interpret specification-sheet parameters such as battery capacity, motor power and rated range. After purchase, a second problem emerges: owners typically have little or no visibility into the ongoing physical condition of the vehicle. This paper presents an e-bike lifecycle platform, organised around a Choose → Monitor → Analyse → Maintain workflow, that addresses both stages. A recommendation module elicits six non-technical requirement inputs (budget, typical distance, terrain, primary use, a stated priority attribute and a charging-convenience preference) and evaluates every entry in a brand-neutral, multi-brand e-bike catalogue using a transparent additive multi-criteria scoring function with fixed, expert-assigned weights. Each ranked result is returned with a percentage match score and a natural-language justification generated from the same computation that produced the score; no deep-learning model has been trained, and the engine is presented as an explainable multi-criteria recommender. A monitoring module has been implemented as an ESP32-based physical sensor node integrated with a web dashboard: a DS18B20 sensor reports motor surface temperature, an MPU6050 accelerometer/gyroscope supports vibration, tilt and possible-movement/impact-pattern detection, and an appropriately rated voltage/current sensor supports battery electrical monitoring, with local buzzer and LED indication and threshold-based status classification and alerts on the dashboard. This paper reports the architecture and integration of this hardware node at the level of what has actually been assembled and connected; it does not report quantitative sensor accuracy, calibration data or long-duration field measurements, which have not yet been collected. To additionally demonstrate multi-unit fleet-level dashboard behaviour beyond the single physical prototype, a parameterised software simulator reproduces State of Charge, State of Health, temperature and position telemetry for a twelve-unit fleet under both nominal and injected-fault conditions; this fleet-scale data is explicitly simulated and is clearly labelled as such throughout. Optional sensing extensions (GPS, BME280 environmental sensing, TPMS tyre-pressure monitoring, an OLED display and solar-assisted charging) remain future work and have not been implemented.

Keywords — e-bike recommendation, multi-criteria decision analysis, explainable recommendation, ESP32, DS18B20, MPU6050, IoT monitoring, battery voltage/current monitoring, fleet telemetry.

I. INTRODUCTION

Electric two-wheeled personal mobility has expanded rapidly in response to rising fuel costs, urban air-quality concerns and improving lithium-ion energy density. A direct consequence of this expansion is product proliferation: a buyer entering the market encounters numerous manufacturers, each publishing specification sheets expressed in engineering units that most non-specialist purchasers cannot easily translate into a purchase decision.

A poorly matched purchase is not a trivial outcome. An e-bike whose rated range, terrain capability or load rating does not correspond to actual usage produces low utilisation, early dissatisfaction and elevated after-sales concern. Selecting among many multi-brand alternatives is therefore a multi-criteria decision problem in which the buyer is typically unable to articulate a formal preference structure.

A second and largely independent problem arises after the transaction. The battery pack, motor and general mechanical condition of an e-bike govern range, service life and rider safety, yet in the mass-market segment the vehicle remains effectively opaque to the owner: overheating, abnormal vibration or electrical irregularities typically pass unobserved until a functional failure occurs.

This paper argues that these two problems – *choosing* an appropriate e-bike and *monitoring* it afterward – are naturally addressed within a single lifecycle-oriented platform, because both are grounded in the same product model. To address the second problem, a small ESP32-based physical sensor node has been built and integrated with a web dashboard, reporting motor surface temperature, vibration/tilt behaviour and battery electrical readings with local and dashboard-level alerting. The motivation for combining an explainable recommendation engine with this IoT-based physical monitoring node is to give a buyer a consistent point of reference across the full ownership lifecycle: **Choose** → **Monitor** → **Analyse** → **Maintain**.

II. RELATED WORK AND RESEARCH GAP

A. Recommender system paradigms

Content-based methods represent items by feature vectors and score candidates by similarity to a profile derived from stated preferences; they operate without a user community and are applicable at launch, but cannot recommend outside the expressed preference region and depend on consistently encoded attribute data. Collaborative filtering exploits the co-occurrence structure of a user–item interaction matrix, with latent-factor formulations achieving strong predictive accuracy on large datasets [1, 2]. Such methods are, however, subject to the cold-start problem: a newly released model or a first-time, non-returning visitor provides no interaction history from which to generalise, which is severe in a retail setting with high catalogue turnover. Hybrid designs combine paradigms [3], and deep-learning recommenders extend representational capacity further [4], at the cost of larger data requirements and reduced interpretability.

B. Multi-criteria decision analysis

Where preferences can be expressed as weighted criteria, multi-attribute decision methods offer an alternative to learned models. The Analytic Hierarchy Process [5] derives criterion weights from pairwise expert comparison, and simple additive weighting produces a scalar utility from normalised criterion

scores. The defining property of this family, for the present application, is auditability: the contribution of each criterion to a final score is inspectable, so a recommendation can be explained in terms a purchaser understands. The corresponding limitation is that weights are fixed a priori and the model does not adapt to observed outcomes.

C. Battery and condition monitoring

State of Charge is not directly measurable and must be inferred; Coulomb counting integrates measured current but accumulates drift, while model-based observers such as the extended Kalman filter applied to equivalent-circuit models [6] correct this by fusing current integration with terminal-voltage prediction. Comparative reviews of electric-vehicle SOC estimation [7] report that filter-based and data-driven estimators substantially outperform open-loop integration over extended horizons. State of Health, expressed as the ratio of present usable capacity to nominal capacity, evolves over hundreds of cycles; reviews of practical SOH estimation [8] distinguish direct measurement, model-based ageing prediction and data-driven regression on cycle-history features. A substantial applied literature separately couples a microcontroller to sensing hardware and a positioning module to transmit telemetry to a remote dashboard for condition monitoring; such systems demonstrate that remote observation reduces unexpected field failure, but are typically vertically integrated with a single product line and validated at small, single-unit scale before any fleet deployment.

D. Research gap

Table 1 summarises the identified gap. Individually, recommendation, multi-criteria decision analysis, battery-state estimation and microcontroller-based instrumented monitoring are each mature and well studied. None of the individual sensing or scoring techniques used in this work is claimed as a novel invention, and no claim is made that this is the first system of its kind. What the reviewed literature does not address is their *integration*: recommendation research rarely optimises for explanation quality toward non-expert buyers, and monitoring research generally assumes single-brand scope and is rarely paired with a purchase-recommendation stage. This paper's contribution is accordingly framed as an integration – of personalised multi-brand recommendation, explainable scoring, and an implemented ESP32-based physical monitoring node with dashboard-level health status and alerts – within one lifecycle-oriented platform.

III. PROPOSED SYSTEM ARCHITECTURE

The system is organised into four layers, with two functionally parallel paths – recommendation and monitoring – that share a single product catalogue.

The *user layer* comprises a prospective purchaser and, on the monitoring side, a dealer or fleet-operator role, both accessing the system through a standard browser without installation or authentication in the present prototype.

The *presentation layer* renders the requirement-input survey and ranked results for the recommendation path (Fig. 1), and gauges, trend charts and an alert list for the monitoring path (Fig. 2).

The *processing layer* contains the two algorithmic components: a scoring engine that evaluates the additive utility function of Section IV and emits a normalised match score with justification strings, and a telemetry processor that applies the classification rules of Section V to derive the active alert set for each unit, regardless of whether the record originates from the physical sensor node or the fleet simulator described below.

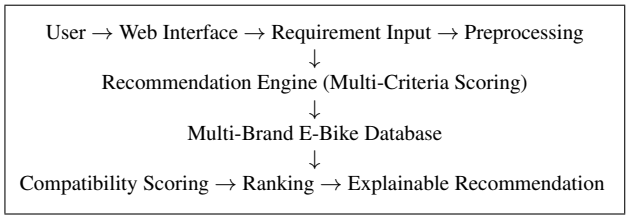


Figure 1. System architecture – recommendation path.

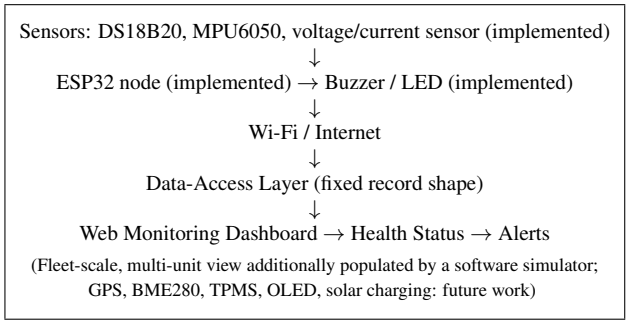


Figure 2. System architecture – monitoring path. The sensor node, ESP32, buzzer/LED and dashboard link are implemented; multi-unit fleet records and the additional listed sensors are not.

The *data layer* holds the brand-neutral catalogue and the telemetry source. The telemetry source exposes a fixed record shape per unit; the implemented ESP32 node populates one such record from live sensor readings, and a software fleet simulator populates additional records to demonstrate multi-unit dashboard behaviour at a scale the single physical prototype does not yet cover.

A. Catalogue representation

Each catalogue entry is a tuple comprising identity, a nominal price interval, rated range, top assisted speed, motor power, battery capacity, rated payload, a set of terrain tags, a set of use-case tags and a comfort-geometry flag. Interval-valued price and set-valued terrain/use-case attributes are deliberate design choices: they permit partial credit in the scoring function rather than forcing binary category membership.

IV. RECOMMENDATION METHODOLOGY

A. Elicitation

The instrument collects six categorical responses, each expressed without engineering units: primary intended use u (commuting, off-road and adventure, fitness and leisure, cargo and delivery); budget band b; typical distance per charge d; dominant terrain t; the single attribute the purchaser most values p (speed, comfort, battery life, load capacity); and a charging-convenience preference c. The design objective was minimal elicitation burden – no free-text entry, no authentication, no numeric input.

Table 1. Summary of the identified research gap

Existing approach	Provides	Gap
Brand-specific tools, retail advice	Guidance within one manufacturer's line	No brand-neutral comparison on common criteria
Latent-factor and deep recommenders	High accuracy from large interaction data	Score without justification; cold start for new models and anonymous users
Specification sheets, comparison sites	Complete parameter data	Assumes engineering literacy; no needs-to-specification mapping
Manufacturer applications	Status for a single vehicle	No consolidated multi-unit view
Instrumented monitoring prototypes	Accurate measured telemetry at single-unit scale	Rarely paired with a purchase-recommendation stage; fleet-scale deployment costly
Reviewed platforms overall	Either purchase guidance or monitoring	No system spanning both lifecycle stages with a real sensor node plus a purchase-recommendation engine

Table 2. Recommendation criteria and fixed weights

Criterion	Weight	Basis
Primary use	35	Category-tag match, partial credit for adjacent category
Budget	25	Normalised price-interval overlap
Range	15	Asymmetric: shortfall penalised, surplus not rewarded
Terrain	15	Explicit tag match, partial credit for mixed-terrain tag
Priority	10	Normalised model property matching stated priority

B. Scoring function

For a catalogue entry m and response vector r, the total score is the additive combination

$$S(m, r) = \sum_{k \in K} w_k \cdot f_k(m, r) \tag{1}$$

where $K = \{\text{use, budget, range, terrain, priority}\}$ and each $f_k \in [0, 1]$ is a criterion satisfaction function. The weight vector is fixed at $w = (35, 25, 15, 15, 10)$, summing to 100 so that S is directly interpretable as a percentage match (Table 2). These weights are fixed and expert-assigned; they are not learned from data, and the scoring function is not a trained machine-learning model.

The use-case criterion admits partial credit for an adjacent category:

$$f_{use} = \begin{cases} 1 & u \in U(m) \\ 15/35 & u = \text{fitness} \wedge \text{commute} \in U(m) \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

The budget criterion measures interval overlap between the model's price interval $P(m)$ and the selected band $B(b)$. Let ω denote the width of the intersection and ω_{min} the width of the narrower interval:

$$f_{budget} = \min\left(1, \frac{\omega}{\omega_{min}}\right) \tag{3}$$

The range criterion is asymmetric:

$$f_{range} = \begin{cases} 1 & R(m) \geq D(d) \\ R(m)/D(d) & \text{otherwise} \end{cases} \tag{4}$$

with the requirement map $D = \{\text{under 20 km} \to 15,\ 20\text{–}40 \to 35,\ 40\text{–}60 \to 55,\ \text{above } 60 \to 65\}$ km.

The terrain criterion grants full credit for an explicit tag match and partial credit for a general mixed-terrain tag:

$$f_{terrain} = \begin{cases} 1 & t \in T(m) \\ 7/15 & \text{mixed} \in T(m) \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

The priority criterion maps the nominated attribute onto a normalised model property, e.g. $f = \min(1, (v(m) - 22)/3.33)$ for speed, where v is top assisted speed in km/h; $f = \min(1, R(m)/70)$ for battery life; and $f = \min(1, C(m)/180)$ for load capacity, where C is rated payload in kilograms. The result is clamped to $[0, 100]$ and rounded. The function is deterministic and stateless.

C. Justification generation

Explanations are emitted as a by-product of scoring rather than reconstructed afterward. Each criterion evaluation that clears a confidence threshold – full use-case match, budget credit of at least 18 of 25 points, satisfied range requirement, explicit terrain match, or priority credit of at least 8 of 10 – appends a corresponding natural-language clause, so the displayed rationale cannot diverge from the arithmetic that produced the score.

V. MONITORING METHODOLOGY

A. Telemetry record and refresh

Each monitored unit exposes a record containing State of Charge (per cent, where applicable), State of Health (per cent, where applicable), temperature (degrees Celsius), a vibration/motion indicator, a charging flag and a derived status label. For the implemented physical node, temperature is read from the DS18B20 sensor and vibration/tilt indication from the MPU6050; battery electrical condition is read from the on-board voltage/current sensor. For the twelve-unit fleet

Table 3. Status classification rules, evaluated in order

Status	Predicate	Interpretation
Critical	$T \geq 50°\text{C}$ or SOC $\leq 8\%$	Thermal protection may engage, or possible risk of immobilisation
Warning	$T \geq 44°\text{C}$ or SOC $\leq 20\%$ or SOH $< 75\%$	Elevated temperature, low reserve, or pack past service threshold
Charging	charging flag set	Replenishing; no threshold violated
Normal	otherwise	All channels within nominal bounds

view used to demonstrate dashboard behaviour at scale, State of Charge, State of Health, temperature and position are instead produced by a software simulator, because only one physical node has been built. The dashboard samples records on a refresh cycle and retains a rolling window of recent samples per channel to provide short-term trend history.

B. Status classification

Status is assigned by ordered evaluation of deterministic threshold rules, given in Table 3. Ordering matters: critical predicates are tested before warning predicates, and the charging state is reported only when no threshold is violated, so an alarm condition is never masked by an in-progress charge. The same rule set is applied uniformly whether the underlying record came from the implemented sensor node or the fleet simulator, since both populate the same fixed record shape.

The SOH warning bound of 75% reflects the conventional end-of-service-life criterion for traction packs. Vibration/tilt readings from the MPU6050 are used to flag a *possible abnormality* or *possible impact/movement condition*; consistent with Section I, MPU6050 data alone does not constitute a definitive accident, theft or mechanical-failure diagnosis, and the dashboard wording is restricted accordingly ("possible abnormality", "condition indication", "inspection recommended").

C. Fleet-scale simulation and fault injection

Because only one physical sensor node has been built, multi-unit fleet-scale dashboard behaviour is additionally demonstrated using a parameterised software simulator, clearly distinguished from the hardware-derived readings described in Section VI. Discharge and charge proceed at bounded random rates; temperature relaxes toward a load-dependent target by a first-order lag with stochastic excursions; SOH decays slowly; and position follows a bounded random walk. A purely nominal simulated fleet exercises none of the alert logic, so the simulator implements persistent fault modes: a thermal fault holds the temperature target in the 48–55°C band; a deep-discharge fault constrains SOC to 2–8%; and a degradation fault pins SOH within 62–74%. All values produced this way are labelled *simulated telemetry* throughout this paper and are never presented as physical sensor measurements.

VI. HARDWARE AND SOFTWARE IMPLEMENTATION

A. Physical sensor node (implemented)

An ESP32 development board forms the controller of a physical sensor node. It is wired to a DS18B20 digital temperature sensor, mounted to read *motor surface temperature* (not internal winding temperature); an MPU6050 six-axis accelerometer/gyroscope for vibration, tilt and motion-pattern sensing; and an appropriately rated voltage/current sensor for battery electrical monitoring, sized for a typical 36 V/48 V e-bike pack. A buzzer and LED indicators are driven from the ESP32's GPIO pins to provide local alert indication in addition to the dashboard alert list. The ESP32 firmware reads these sensors, applies the status-classification logic of Table 3, and transmits the resulting record over Wi-Fi to the data-access layer, which the same web dashboard used for the fleet view consumes. This node has been assembled and connected end-to-end: the sensors, the ESP32, the buzzer/LED alerting and the Wi-Fi link to the dashboard are implemented. Quantitative sensor accuracy, calibration coefficients and long-duration or load-condition measurements have not yet been collected and are not reported in this paper (Section VIII).

Optional sensing modules – GPS for location/anti-theft tracking, a BME280 environmental sensor for ambient temperature, humidity and atmospheric (not tyre) pressure, a dedicated TPMS sensor for actual tyre-pressure monitoring, an OLED status display, and a solar-assisted charging module with charging-energy monitoring – are not part of the implemented node and remain future work (Section IX).

B. Software stack

The web dashboard and recommendation interface are implemented as a static client-side application in HTML5, CSS3 and ECMAScript 2015, with no build step, server-side component, database or authentication service. The data-access layer that receives the ESP32 node's transmitted records and the fleet simulator's generated records exposes a single fixed record shape to the presentation layer, so the dashboard code is agnostic to which source populated a given record. No React, Node.js, MongoDB, Flask, FastAPI, Python-based machine-learning model or cloud database is used in the current implementation. Cartography, where used for fleet-position display, uses the Leaflet library [9] against OpenStreetMap raster tiles [10].

The measured software artefact comprises a catalogue of eight models spanning commuter, folding, off-road, cargo, fitness and comfort categories, one implemented physical sensor node, and a simulated fleet of twelve additional units (four carrying persistent faults) used to demonstrate multi-unit dashboard behaviour.

VII. RESULTS AND DISCUSSION

A. Traced scoring decompositions

Because the scoring function is deterministic, its behaviour is reported by tracing complete decompositions rather than by summary statistics. Table 4 traces two contrasting response vectors through Eqs. (1)–(5), with per-criterion contributions shown in points.

Table 4. Score decomposition for two response vectors

Criterion	Profile A: commuter	Profile B: cargo
Response vector	commute; under Rs. 50k; 20–40 km; flat city; speed	cargo; above Rs. 1.5L; above 60 km; off-road; load capacity
Top-ranked model	Urban Glide (city commuter)	Cargo Hauler (cargo / delivery)
Use case (35)	35 – exact tag match	35 – exact tag match
Budget (25)	13 – partial interval overlap	0 – price interval below the selected band
Range (15)	15 – 40 km $\geq$ 35 km required	14 – 60 km against 65 km required
Terrain (15)	15 – explicit city tag	7 – mixed-terrain tag only
Priority (10)	9 – 25 km/h top speed	10 – 180 kg rated payload
Total S	**87**	**66**

Both traces reproduce the values observed in the deployed application, confirming that the implementation realises the specification. In Profile A the selected model is penalised only on budget; the resulting score of 87 signals a strong but imperfect match. Profile B is more instructive: the cargo requirement and payload are matched exactly, but the buyer's stated budget lies entirely above the appropriate product's price interval, and the requested off-road capability is only partially met. The score of 66 encodes a genuine tension in the request rather than a deficiency in the product. Equation (3) treats over-budget and under-budget mismatch symmetrically; asymmetric treatment is identified as future work (Section IX).

B. Physical sensor node integration

The implemented ESP32 node was connected end-to-end: the DS18B20, MPU6050 and voltage/current sensor were wired to the board, the firmware was flashed and executed, and the resulting records were observed to reach the web dashboard over Wi-Fi, where the threshold rules of Table 3 were confirmed to classify status and drive the buzzer/LED and dashboard alert list from live sensor input. This establishes that the hardware-to-dashboard pipeline of Fig. 2 functions as designed. No specific numeric temperature, vibration or electrical readings, sensor accuracy figures, or calibration results are reported here, because dedicated bench characterisation of the sensor node has not yet been carried out; this is stated explicitly rather than omitted, and is listed as a limitation in Section VIII.

C. Fleet-scale simulated dashboard behaviour

Table 5 reports the status distribution of the twelve-unit *simulated* fleet, observed on load and after approximately twenty seconds of continuous simulation, demonstrating the effect of the persistent fault design of Section V-C. This subsection concerns the software fleet simulator only, used to exercise the dashboard at a multi-unit scale the single physical node does not yet provide; no physical unit contributed to these figures.

The deep-discharge-faulted unit held SOC at 5% throughout, and the two thermal-faulted units settled at 49–52°C, crossing the 50°C critical bound intermittently, showing that the dashboard tracks boundary conditions rather than latching a single state. These figures characterise the simulator, not the physical sensor node.

Table 5. Observed status distribution, 12-unit simulated fleet

Status	On load	After $\sim$20s	Notes
Normal	8	6	Unfaulted units remain nominal
Warning	2	3	Includes degradation-faulted unit, held at 69% SOH
Critical	1	2	Thermal-faulted units oscillate 49–52°C across the 50°C bound
Charging	1	1	Reported only when no threshold is violated

D. Scope of the present evaluation

The results above establish that (i) the recommendation scoring specification is correctly realised in the deployed application, (ii) the physical ESP32 sensor node has been built and successfully delivers live readings to the dashboard end-to-end, and (iii) the dashboard correctly classifies and displays both live and simulated multi-unit telemetry using one shared rule set. The results do not constitute an evaluation of recommendation quality or of sensor accuracy: no user study has been conducted, no ground-truth suitability labels exist for the catalogue, the recommendation weights have not been empirically validated against purchaser outcomes, and the physical sensor node has not been bench-tested against a reference instrument. No accuracy, precision, recall, F1 score or comparable performance metric is reported for either module, because none has been measured.

VIII. LIMITATIONS

Several limitations bound the claims of this paper. First, only one physical sensor node has been built; fleet-scale physical deployment has not been attempted, and the twelve-unit fleet view remains software-simulated. Second, the DS18B20, MPU6050 and voltage/current sensor readings from the implemented node have not been bench-characterised against a reference instrument, so no sensor accuracy, precision or calibration figure is reported, and the status thresholds of Table 3 –

inherited from the simulated-fleet design – have not been independently validated against the physical sensor node's real operating range. Third, MPU6050-based vibration/tilt sensing supports only possible-abnormality indication, not a validated accident, theft or mechanical-failure diagnosis. Fourth, the recommendation criterion weights are fixed and expert-assigned; the system does not learn from user selections, and no sensitivity analysis of ranking stability has been performed. Fifth, the catalogue comprises eight illustrative entries with placeholder commercial attributes and does not represent the full real market. Sixth, recommendation quality has not been measured against any external criterion, so no accuracy claim is advanced. Seventh, the implemented node depends on Wi-Fi connectivity, and any future map-based fleet view depends on the public OpenStreetMap tile service, provisioned for light and development use.

IX. FUTURE WORK

A. Optional sensing extensions

GPS-based location/anti-theft tracking, a BME280 environmental sensor for ambient temperature, humidity and atmospheric pressure, a dedicated TPMS tyre-pressure sensor, an OLED local status display, and solar-assisted battery charging with charging-energy monitoring are proposed extensions to the implemented node. None of these has been implemented; none of these individual sensing technologies is claimed as novel, and their value lies in integration with the existing pipeline of Fig. 2 once added.

B. Validation of the physical sensor node

The immediate next step is bench characterisation of the implemented DS18B20, MPU6050 and voltage/current sensor readings against reference instruments, recalibration of the status thresholds of Table 3 against the resulting data, and deployment of additional physical nodes to replace the software fleet simulator with a genuinely multi-unit physical fleet. A fleet-level health score derived from the combined sensor channels is proposed as a prototype decision/condition indicator – explicitly not a clinically or industrially validated battery-health estimator – to generate maintenance recommendations of the form: abnormal temperature $\rightarrow$ overheating-related warning $\rightarrow$ inspection recommended; abnormal electrical behaviour $\rightarrow$ possible electrical/battery warning $\rightarrow$ inspection recommended; abnormal vibration/movement $\rightarrow$ possible mechanical abnormality $\rightarrow$ inspection recommended; repeated abnormal patterns $\rightarrow$ maintenance recommendation. This scoring has not yet been implemented or validated.

C. Recommendation enhancements

The budget criterion should be made asymmetric, penalising exceeding the nominated band more lightly than falling below it. The fixed weight vector should be subjected to sensitivity analysis and, given logged selection data, refined by learning-to-rank methods so the model adapts while retaining an explainable additive structure; this is a proposed architectural extension only, not implemented. A controlled user study comparing recommendation acceptance against unaided selection would provide quality evidence the present work does not claim.

D. Closing the feedback loop

Longer term, real-world usage data collected through the physical sensor node (once deployed at fleet scale) could be fed back into the recommendation engine to compare predicted suitability against observed in-service performance, and accumulated telemetry would enable predictive rather than reactive maintenance using data-driven remaining-useful-life methods such as those surveyed in [8]. This feedback loop is a proposed direction only; it has not been experimentally validated, and no such data currently exists.

X. CONCLUSION

This paper has presented an e-bike lifecycle platform – Choose $\rightarrow$ Monitor $\rightarrow$ Analyse $\rightarrow$ Maintain – that connects a pre-purchase, explainable multi-criteria recommendation engine with a post-purchase condition-monitoring system. The recommendation module reduces elicitation to six non-technical inputs and scores candidates with a transparent additive function whose per-criterion contributions were traced explicitly; no deep-learning model is claimed. For monitoring, an ESP32-based physical sensor node – integrating a DS18B20 motor surface-temperature sensor, an MPU6050 for vibration/tilt/possible-movement sensing, an appropriately rated voltage/current sensor, and local buzzer/LED alerting – has been implemented and connected end-to-end to a web dashboard that applies deterministic threshold-based status classification and alerting. A software fleet simulator additionally demonstrates the same dashboard logic at multi-unit scale, and is clearly distinguished throughout from the physical node's live data.

The contribution is best characterised as a working recommendation engine together with a genuinely implemented, but not yet bench-validated or fleet-scaled, physical monitoring node – rather than as an empirically validated recommender or a fully validated IoT product. Quantitative sensor calibration and accuracy testing, evaluation of recommendation quality, and physical fleet-scale deployment remain necessary and are identified as the next stage of this work. No claim of commercial readiness, worldwide novelty, trained machine-learning models, or large-scale field validation is made.

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Note to authors: volume, issue, page and DOI fields for references [1]–[8] should be verified against the primary sources before final submission. If bench-characterisation readings for the DS18B20/MPU6050/voltage-current sensor become available before your viva, they should be added to Section VII-B with actual measured values – do not let anyone (including an AI assistant) substitute invented numbers for them.